COLLEGE NAME : DON BOSCO COLLEGE (CO-ED), YELAGIRI HLLS

COLLEGE CODE : 358

DEPARTENT : B.SC(COMPUTER SCIENCE)

TITLE : COURIER MANAGEMENT SYSTEM (CMS)

TEAM MEMBERS NAME :

1. **PRIYADHARSHINI.K - (35823U18038)**
2. **RAMYA.R – (35823U18039)**
3. **ROBIN PIOUS.V - (35823U18040)**
4. **ROZRIO.X - (35823U1841)**
5. **RUTH EPSIBA.A - (35823U18042)**

ABSTRACT

Courier Management System (CMS) plays an important role in improving logistics operations, customer service, and delivery efficiency. The project titled **"A Courier Management System (CMS)"** focuses on designing and implementing a cloud-based application using the Salesforce platform to automate courier booking, tracking, and delivery management processes.Courier service companies handle numerous parcel bookings and deliveries every day, including both local and long-distance shipments. Previously, shipment details, customer records, and delivery status were maintained manually, which often led to errors, delays in parcel tracking, and inefficient record management.

To overcome these challenges, a centralized Courier Management System was developed. The application includes custom objects such as **Sender Details, Receiver Details, Courier Booking, and Delivery Status** to store and manage courier information in a structured manner. Formula fields are used to automatically calculate **shipping charges and delivery estimates** based on parcel weight, distance, and service type, thereby reducing manual calculation errors.

The system also incorporates **validation rules** to ensure accurate data entry. These rules help prevent incorrect phone numbers, address details, and identification numbers from being saved in the system, maintaining high data quality and reliability.

Record types and customized page layouts are created to differentiate between **standard delivery and express delivery services**, providing a user-friendly interface and better categorization of courier records.

An important feature of the system is **automation through scheduled and triggered flows**, which automatically update delivery status, notify customers about shipment progress, and send alerts for pending deliveries.

Additionally, **reports and dashboards** are generated to provide real-time graphical insights into shipment volumes, delivery performance, and customer activity. These analytics help managers make informed and data-driven decisions to improve service efficiency.

The Courier Management System is **cloud-based, secure, scalable, and accessible 24/7**, ensuring high availability and reliability. Overall, this project demonstrates the practical implementation of logistics management concepts using Salesforce and highlights how automation and analytics can significantly improve operational efficiency, data accuracy, and customer satisfaction in courier services.

INTRODUCTION

In today's fast-growing logistics and delivery industry, managing parcel records and delivery information efficiently is essential for providing reliable customer service. Many courier companies still depend on manual record-keeping methods such as paper registers or spreadsheets. These traditional methods often lead to data entry errors, misplaced shipment details, delays in tracking parcels, and difficulty in generating delivery reports.

Courier service businesses handle numerous parcel bookings and deliveries every day, including both local and long-distance shipments. As the volume of shipments increases, managing sender details, receiver information, parcel tracking, and delivery updates manually becomes challenging. Employees may spend more time maintaining records instead of focusing on improving delivery services and customer satisfaction.

To address these challenges, a **cloud-based Courier Management System (CMS)** was developed using the **Salesforce platform**. Salesforce provides a powerful environment for building customized applications that support automation, secure data storage, and real-time reporting.

The **Courier Management System** developed includes:

- **Custom objects** to manage Sender Details, Receiver Details, and Courier Booking.
- **Formula fields** to automatically calculate shipping charges.
- **Validation rules** to ensure accurate data entry.
- **Reports and dashboards** to monitor delivery performance and shipment records.

By implementing this system, the courier company can centralize shipment data, reduce manual work, improve accuracy, and provide faster parcel tracking services. The system also helps managers make better decisions using real-time reports.

This project demonstrates how technology can convert traditional courier operations into an efficient and automated digital system.

PROBLEM STATEMENT

The courier company currently lacks a centralized and automated system to manage parcel bookings, shipment tracking, and delivery records efficiently. The existing process depends on manual data entry and shipping charge calculations, which leads to operational inefficiencies and errors.

Problems Faced by Courier Staff:

Courier staff are responsible for managing parcel bookings and generating shipment records. However:

- Parcel weight and shipping charges are calculated manually.
- Manual calculations often result in incorrect billing or shipment charges.
- Sender and receiver details must be updated manually without automation support.

Because of these issues, courier staff experience stress and inefficiency while handling daily parcel bookings.

Problems Faced by Operations Managers

Operations managers need accurate reports to monitor courier deliveries and shipment performance. However:

- Delivery reports are delayed and sometimes incomplete.
- Shipment data is stored in multiple files and registers.
- There is no centralized system for real-time tracking and analysis.

This makes decision-making difficult and reduces overall operational efficiency.

Problems Faced by Customers

Customers expect quick parcel tracking and timely delivery updates. However:

- There is no automated notification system.
- Customers do not receive shipment status updates.
- Customer communication and engagement are minimal.

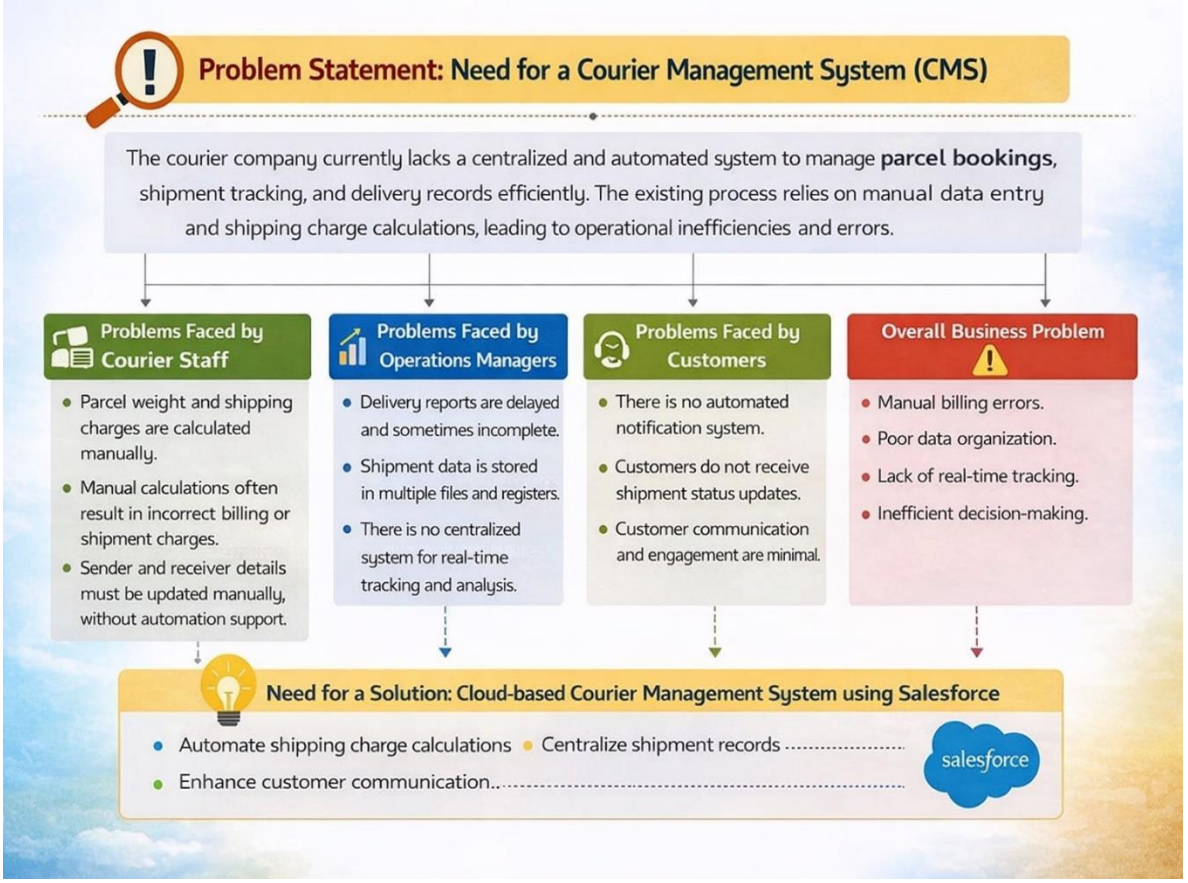
Overall Business Problem

Due to the absence of an automated Courier Management System (CMS), the courier company faces:

- Manual shipping charge calculation errors
- Poor shipment data organization.
- Lack of real-time parcel tracking
- Reduced customer communication.
- Inefficient operational decision-making

Additionally, the company lacks proper data validation mechanisms to ensure accurate shipment information entry. The absence of standardized data fields leads to inconsistencies in sender and receiver records. There is no structured workflow to streamline parcel booking and delivery tracking activities. These operational gaps directly impact service efficiency, delivery accuracy, and overall business growth.

Therefore, the organization requires a cloud-based Courier Management System using Salesforce to streamline courier operations, automate shipping charge calculations, centralize shipment data management, and improve parcel tracking and customer communication**.**



PROPOSED SOLUTION

To overcome the challenges faced by the courier company, a **cloud-based Courier Management System (CMS)** is proposed and developed using **Salesforce**. The proposed system provides a **centralized, automated, and structured solution** for managing parcel bookings, shipment tracking, and delivery reporting.

Centralized Data Management

The system introduces custom objects such as:

- Sender Details
- Receiver Details
- Courier Booking

These objects help in storing and organizing courier data in a **structured and centralized manner**.

Automation of Shipping Charges

To reduce manual errors:

- Formula fields are created to automatically calculate parcel weight and shipping charges.
- Formula fields generate the total delivery cost automatically.

This ensures accurate billing without manual calculations and reduces workload for courier staff.

Data Validation Mechanism

Validation rules are implemented to:

- Ensure correct format for phone numbers.
- Validate PIN code format.
- Restrict incorrect address or shipment details.

This improves data accuracy and prevents incorrect shipment information from being saved.

Record Types and Page Layout Customization:

Two record types are created:

- Standard Delivery
- Express Delivery
- Same-Day Delivery

Separate page layouts are designed to provide a better user interface and proper classification of courier shipment records. This helps staff easily enter and manage parcel details based on the delivery type.

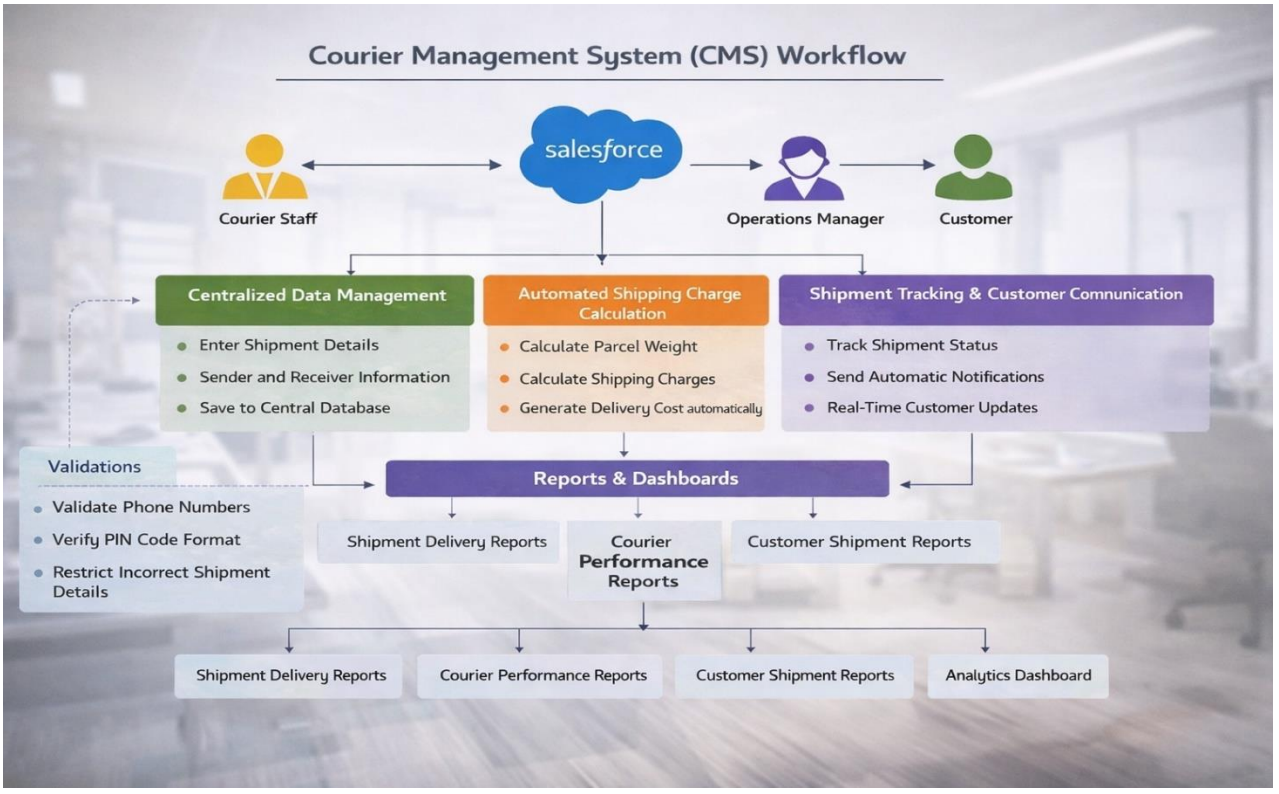
Reporting and Dashboard Analytics

The system generates:

- Shipment Delivery Reports.
- Courier Performance Reports.
- ustomer Shipment Reports.
- Interactive Dashboards.

These reports provide real-time graphical insights that help managers monitor shipment status, analyze delivery performance, and make informed operational decisions.

WORK FLOW DIAGRAM



SUMMARY

The project **"Courier Management System (CMS)"** focuses on developing a cloud-based management system using the Salesforce platform to automate courier service operations and shipment management processes.

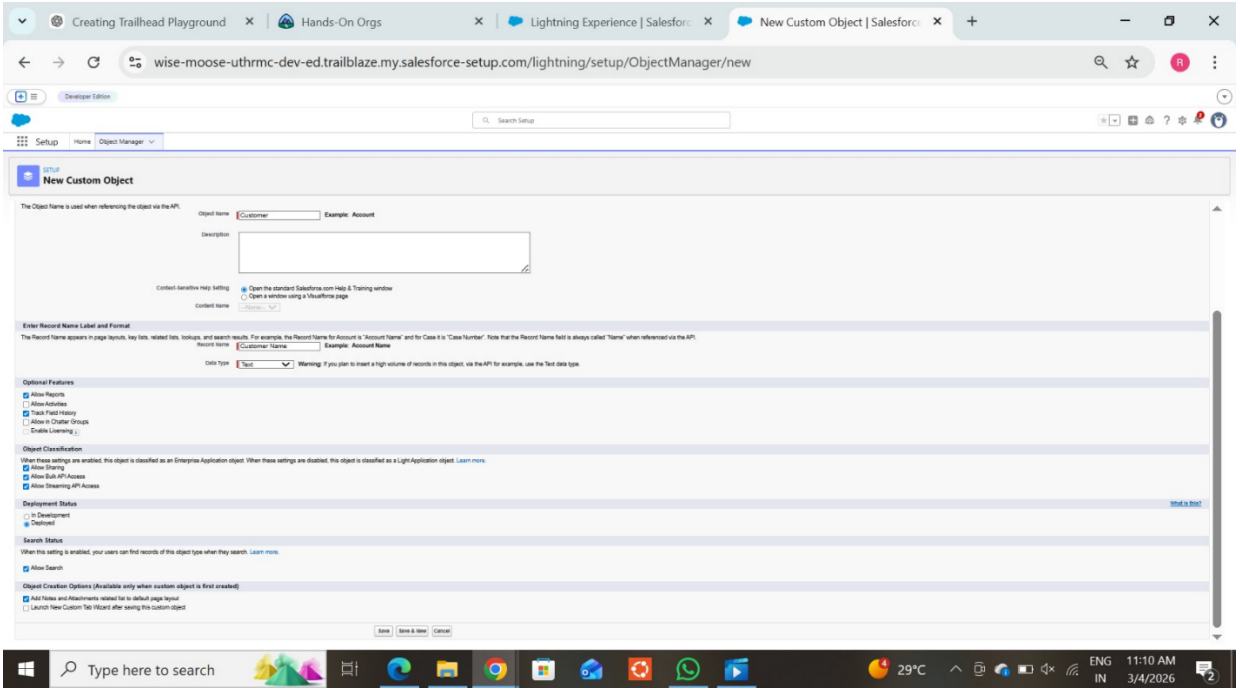
The system replaces manual record-keeping and shipment calculations with a centralized and automated solution. Custom objects such as **Sender Account, Receiver Account, and Shipment Details** were created to ensure structured data storage. Formula fields automatically calculate shipment charges based on weight, distance, and delivery type, improving billing accuracy and reducing human errors.

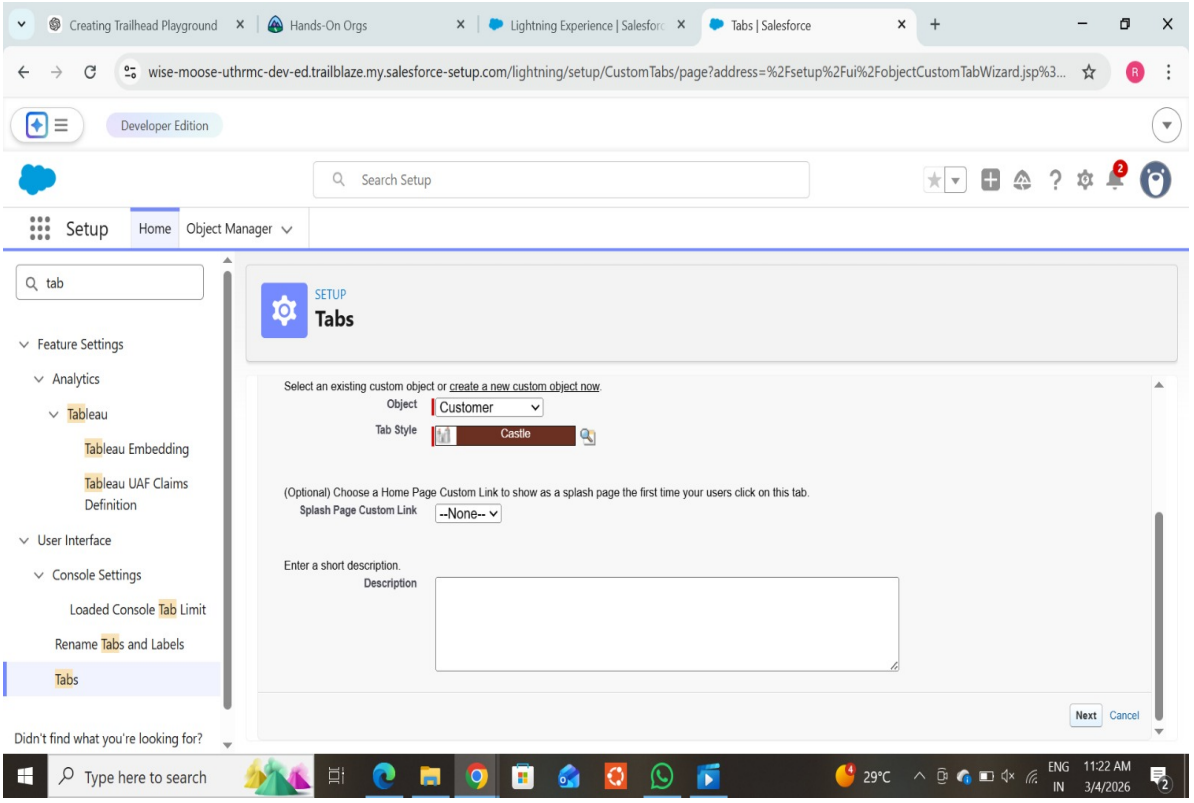
Validation rules were implemented to maintain proper data entry formats for phone numbers, PIN codes, and address details. Record types were introduced to differentiate between **Domestic Shipments** and **International Shipments**, improving shipment classification and management. A schedule-triggered flow was configured to automatically send delivery notifications and shipment status updates to customers, strengthening customer communication and service reliability.

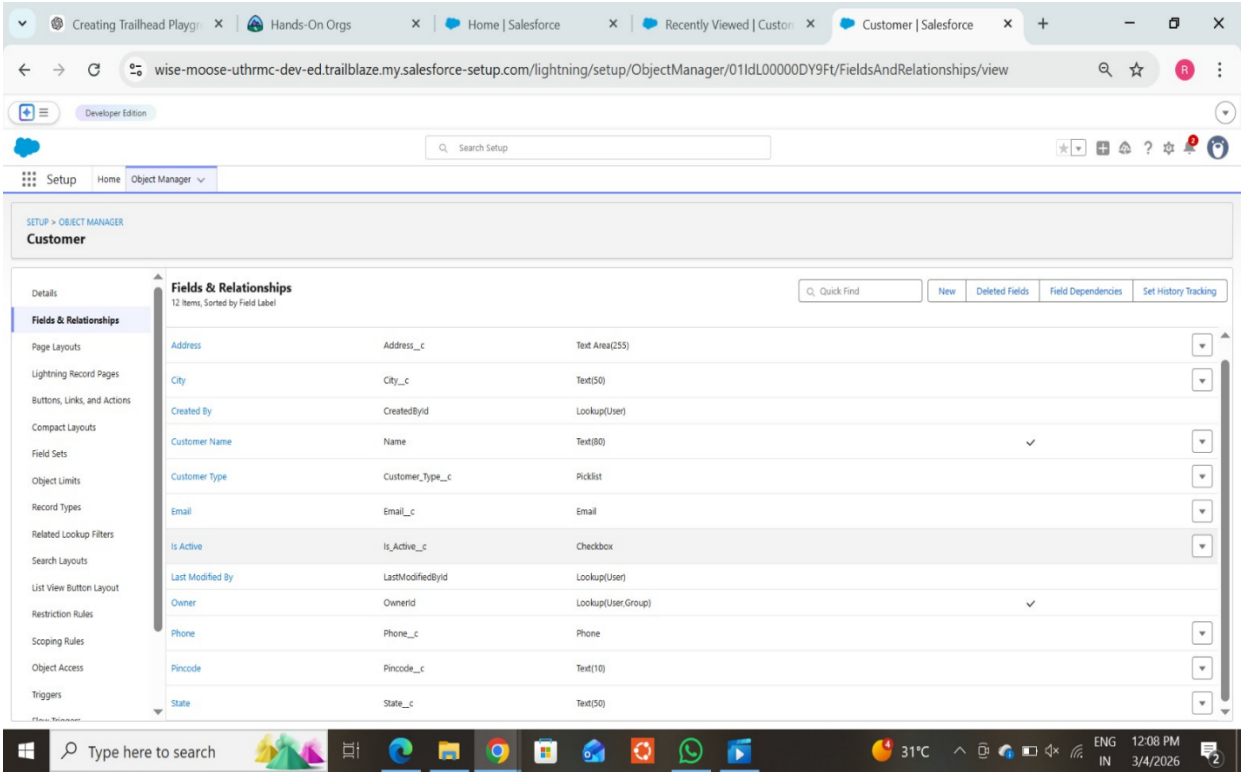
Reports and dashboards were designed to provide real-time insights into shipment status, delivery performance, and business activities. These analytical tools assist managers in monitoring courier operations and making informed decisions. The centralized database ensures easy access, better organization, and improved transparency in shipment handling.
Furthermore, the system enhances operational efficiency by minimizing repetitive manual tasks and reducing processing time. The cloud-based environment ensures data security, reliability, and 24/7 accessibility. The application is scalable, allowing courier services to expand operations without major structural changes.

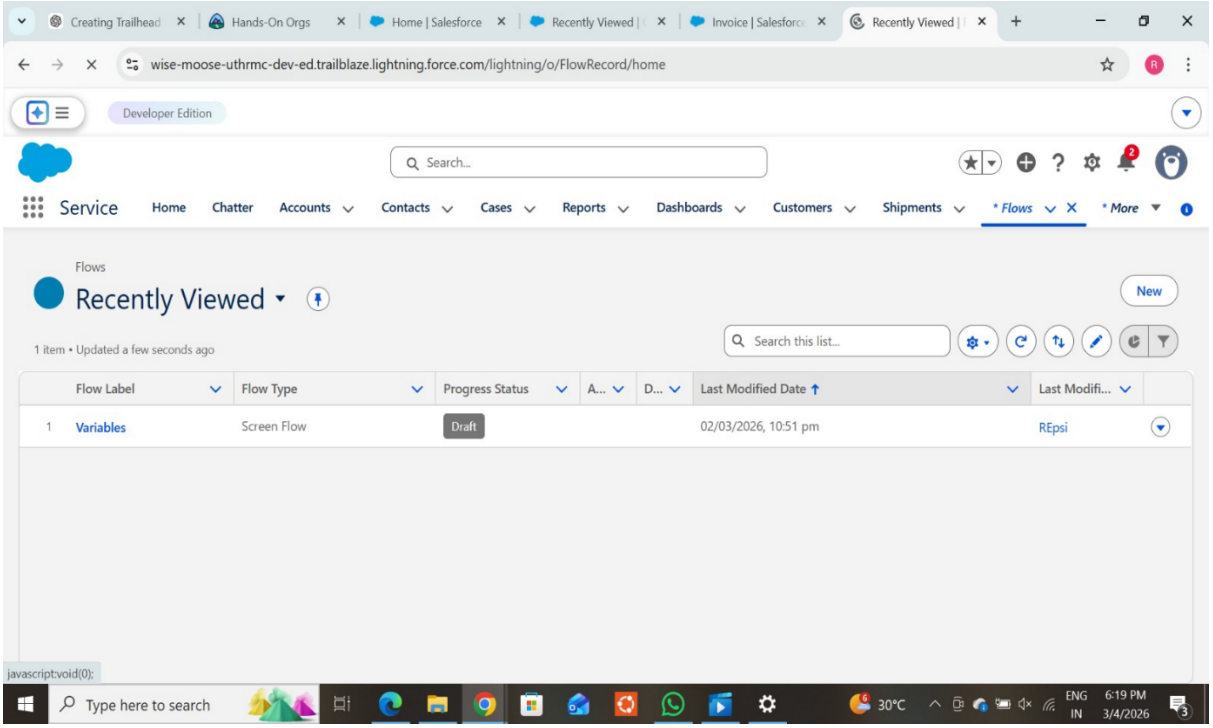
The implementation of automation features reduces dependency on manual supervision. The CMS improves coordination between delivery staff and management. It enables faster tracking of courier shipments and quick retrieval of sender and receiver information. The platform supports accurate tracking of shipment history and delivery status. The structured workflow ensures smooth courier booking, processing, and delivery management. Overall, the system contributes to improved service quality and efficient courier operations.

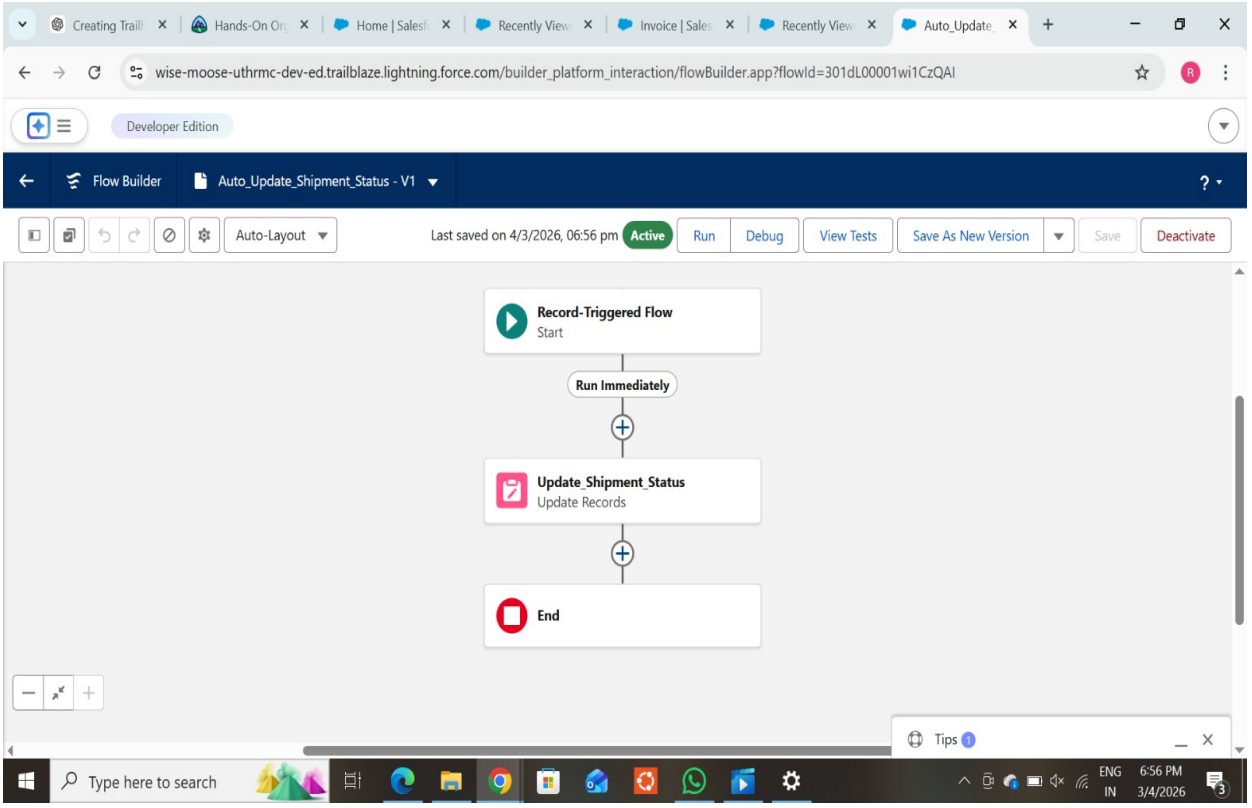
SCREEN SHORTS

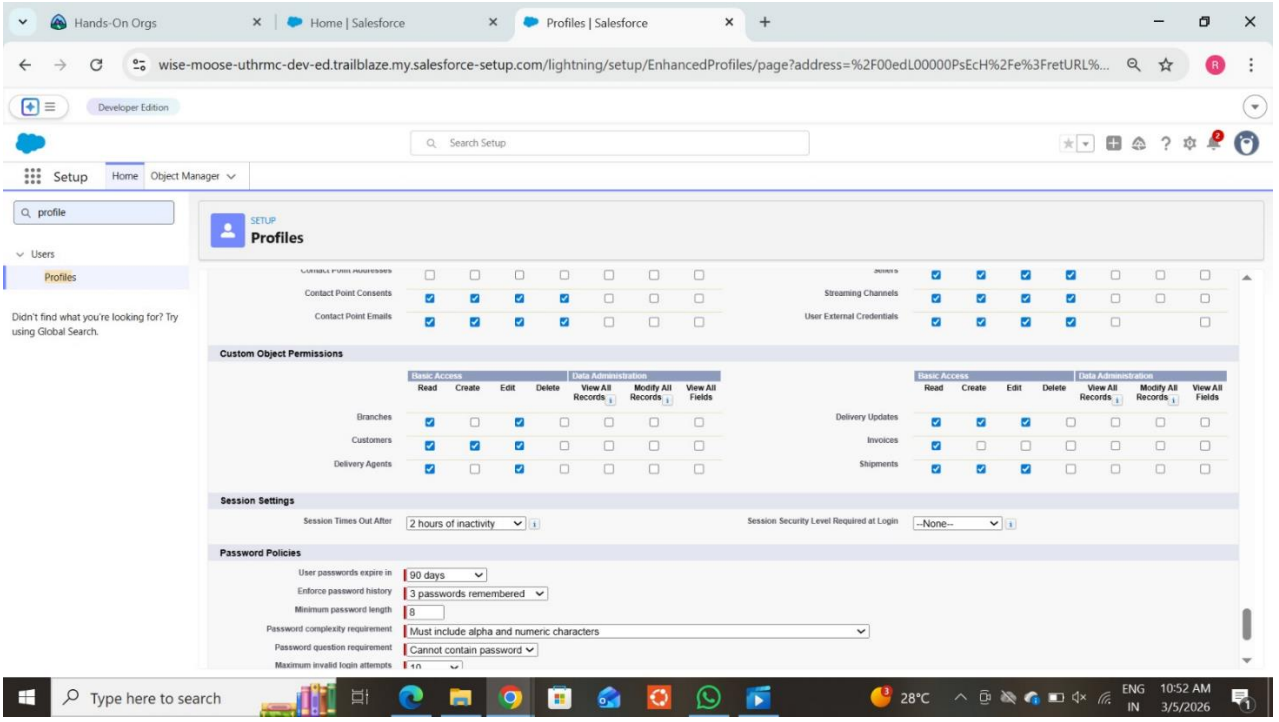





Creating Trailhead Playgr
Hands-On Orgs
Home | Salesforce
Recently Viewed | Custon
Customer | Salesforce
wise-moose-uthrmc-dev-ed.trailblaze.my.salesforce-setup.com/lightning/setup/ObjectManager/01IdL00000DY9Ft/FieldsAndRelationships/view
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Customer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Fields & Relationships
12 items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
Set History Tracking
Address Address__c Text Area(255)
City City__c Text(50)
Created By CreatedById Lookup(User)
Customer Name Name Text(80)
Customer Type Customer_Type__c Picklist
Email Email__c Email
Is Active Is_Active__c Checkbox
Last Modified By LastModifiedById Lookup(User)
Owner OwnerId Lookup(User,Group)
Phone Phone__c Phone
Pincode Pincode__c Text(10)
State State__c Text(50)
Type here to search
31°C
ENG IN
12:08 PM
3/4/2026


Creating Trailhead
Hands-On Orgs
Home | Salesforce
Recently Viewed |
Invoice | Salesforce
Recently Viewed |
wise-moose-uthrmc-dev-ed.trailblaze.lightning.force.com/lightning/o/FlowRecord/home
Developer Edition
Search...
Service
Home
Chatter
Accounts
Contacts
Cases
Reports
Dashboards
Customers
Shipments
* Flows
* More
Flows
Recently Viewed
New
1 item • Updated a few seconds ago
Search this list...
Flow Label
Flow Type
Progress Status
A...
D...
Last Modified Date
Last Modifi...
1 Variables Screen Flow Draft 02/03/2026, 10:51 pm REpsi
javascript:void(0);
Type here to search
30°C
ENG IN
6:19 PM
3/4/2026


Creating Traill
Hands-On Or
Home | Salesf
Recently View
Invoice | Sales
Recently View
Auto_Update
wise-moose-uthrmc-dev-ed.trailblaze.lightning.force.com/builder_platform_interaction/flowBuilder.app?flowId=301dL00001wi1CzQAI
Developer Edition
Flow Builder
Auto_Update_Shipment_Status - V1
Auto-Layout
Last saved on 4/3/2026, 06:56 pm
Active
Run
Debug
View Tests
Save As New Version
Save
Deactivate
Record-Triggered Flow
Start
Run Immediately
Update_Shipment_Status
Update Records
End
Tips
Type here to search
ENG
IN
6:56 PM
3/4/2026


Hands-On Orgs
Home | Salesforce
Profiles | Salesforce
wise-moose-uthrmc-dev-ed.trailblaze.my.salesforce-setup.com/lightning/setup/EnhancedProfiles/page?address=%2F00edL00000PsEcH%2Fe%3FretURL%...
Developer Edition
Search Setup
Setup
Home
Object Manager
profile
Users
Profiles
Didn't find what you're looking for? Try using Global Search.
SETUP
Profiles
Contact Point Consents
Contact Point Emails
Streaming Channels
User External Credentials
Custom Object Permissions
Basic Access
Data Administration
Read
Create
Edit
Delete
View All Records
Modify All Records
View All Fields
Branches
Customers
Delivery Agents
Delivery Updates
Invoices
Shipments
Session Settings
Session Times Out After
2 hours of inactivity
Session Security Level Required at Login
--None--
Password Policies
User passwords expire in
90 days
Enforce password history
3 passwords remembered
Minimum password length
8
Password complexity requirement
Must include alpha and numeric characters
Password question requirement
Cannot contain password
Type here to search
28°C
ENG
IN
10:52 AM
3/5/2026

Hands-On Orgs | Home | Salesforce | Roles | Salesforce

wise-moose-uthrmc-dev-ed.trailblaze.my.salesforce-setup.com/lightning/setup/Roles/home

Developer Edition

Search Setup

Setup | Home | Object Manager

role

- Users
  - Roles
- Feature Settings
  - Sales
    - Contact Roles on Contracts
    - Contact Roles on Opportunities
  - Service
    - Case Teams
      - Case Team Roles
    - Contact Roles on Cases

Didn't find what you're looking for? Try using Global Search.

SETUP

Roles

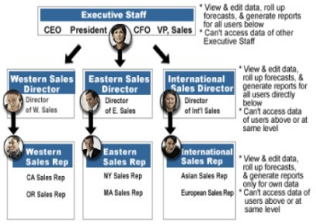
Understanding Roles

Help for this Page

Set up your Role Hierarchy to control how your organization reports on and accesses data.

Sample Role Hierarchy

View other sample Role Hierarchies: Territory-based Sample



Set Up Roles

Don't show this page again

Type here to search | TCS | ENG IN | 11:11 AM 3/5/2026

Hands-On Orgs | Home | Salesforce | Roles | Salesforce

wise-moose-uthrmc-dev-ed.trailblaze.my.salesforce-setup.com/lightning/setup/Roles/home

Developer Edition

Search Setup

Setup | Home | Object Manager

role

- Users
  - Roles
- Feature Settings
  - Sales
    - Contact Roles on Contracts
    - Contact Roles on Opportunities
  - Service
    - Case Teams
      - Case Team Roles
    - Contact Roles on Cases

Didn't find what you're looking for? Try using Global Search.

SETUP

Roles

Creating the Role Hierarchy

Help for this Page

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy | Show in tree view

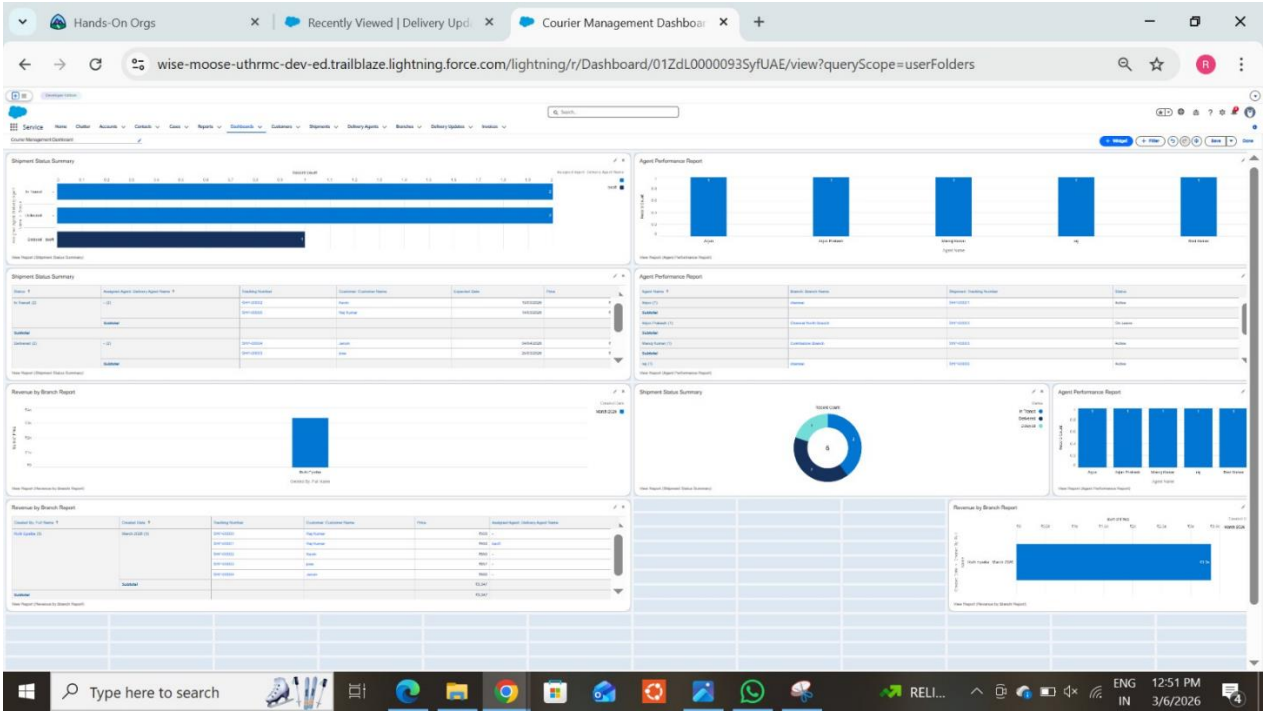
Collapse All | Expand All

- dbcy
  - Add Role
  - CEO Edit | Del | Assign
    - Add Role
    - CFO Edit | Del | Assign
      - Add Role
    - COO Edit | Del | Assign
      - Add Role
    - SVP, Customer Service & Support Edit | Del | Assign
      - Add Role
      - Customer Support, International Edit | Del | Assign
        - Add Role
      - Customer Support, North America Edit | Del | Assign
        - Add Role
      - Installation & Repair Services Edit | Del | Assign
        - Add Role
    - SVP, Human Resources Edit | Del | Assign
      - Add Role
    - SVP, Sales & Marketing Edit | Del | Assign
      - Add Role

javascript:srcUp(%27%2F00E%2Fe%3Fparent%3D00EdL00000MNk5V%26setupid%3DRoles%26retURL%3D%252Fui%252Fsetup%252Fuser%252FRoleViewPage%253Fsetupid%253DRoles%2526isdtp%253Dp1%26isdtp%3Dp1%27);

Type here to search | TCS | ENG IN | 11:12 AM 3/5/2026

OUTPUT:


Hands-On Orgs
Recently Viewed | Delivery Upd
Courier Management Dashboa
wise-moose-uthrmc-dev-ed.trailblaze.lightning.force.com/lightning/r/Dashboard/01ZdL0000093SyfUAE/view?queryScope=userFolders
Shipment Status Summary
Agent Performance Report
Revenue by Branch Report
Type here to search
ENG
IN
12:51 PM
3/6/2026

CONCLUSION

The project “Courier Management System (CMS)” successfully demonstrates the design and implementation of a cloud-based courier service management system using the Salesforce platform. The system effectively resolves the challenges associated with manual shipment records, delivery tracking issues, scattered data, and delayed reporting.

By introducing custom objects such as Sender, Receiver, and Shipment Details, the application ensures structured and centralized data storage. The use of formula fields for automatic shipment charge calculation based on weight and delivery type significantly reduces manual effort and minimizes calculation errors. Validation rules improve data accuracy by enforcing proper input formats for important fields such as phone numbers, PIN codes, and address details.

The implementation of record types and customized page layouts enhances shipment classification and improves user experience. Automated notifications through scheduled flows keep customers informed about shipment status and delivery updates, improving communication and customer satisfaction. Reports and dashboards provide real-time insights into courier activities, shipment performance, and delivery status, enabling managers to monitor operations and make better decisions.

The Courier Management System increases productivity by reducing repetitive manual tasks and improving workflow efficiency. The cloud-based architecture ensures data security, reliability, scalability, and 24/7 accessibility. The integration of automation, reporting, and shipment tracking features into a single platform improves coordination between staff and management while ensuring efficient courier operations.

Overall, the project represents a successful transformation from manual courier management to an automated and intelligent system. It demonstrates practical knowledge of Salesforce tools and highlights how technology-driven solutions can enhance operational efficiency, improve service quality, and support the growth of courier service organizations.